the help of the carburettor and sent into the engine. The air-fuel ratio won't be same for all the time. It should be varied in different ratio according to the needs. For example (i)starting, (ii)slow speed, (iii)idling, (iv) ordinary speed, (v)high speed, (vi)cold or hot climate starting engine are the conditions at which the air-fuel ratio is to be varied and can be varied by three stages. Namely,

- Rich mixture
- Chemically correct (or) stoichiometric mixture
- Lean mixture

9.4.1 Rich Mixture

To burn 1 kg of fuel petrol completely, nearly 15 kg of air is being supplied. If the fuel in the air-fuel mixture is 1kg and air is 15kg it is said to be rich mixture.

Example, 10:1 (10 % air and 1 % fuel). Form the example, the mixture has high burning capacity. At the below given conditions the engine needs rich mixture.

- During starting (A:F = 5:1)
- Idling (A:F=10:1)
- While overtaking (A:F =12:1)

9.4.2 Chemically Correct (or) Stoichiometric Mixture

In the air-fuel mixture if the air is more and petrol is less, then it is called Chemically correct (or) stoichiometric mixture.

Example, 16:1 (16 % of air and 1% of fuel)

In normal and ordinary speed the engine is being operated at normal mixture.

9.4.3 Lean Mixture

If the fuel exceeds beyond the stoichiometric limit, then it is called lean mixture. Example: (18:1). The vehicle will operates at lean mixture in case of low load and high speed.

9.5 CARBURETTOR

The carburettor is the important component in the petrol engine. It makes the liquid form of petrol in to vapour form and mixes it with required amount of air and sends into the cylinder. It also atomises the fuel for easy mixing with air. This process is called carburetion.

9.5.1 Function of Carburettor

1. It stores the required amount of petrol in the float chamber
2. It vaporizes the fuel and mixes with air
3. Carburetion is done as per engine load and speed
4. Rich mixture is supplied to engine during starting and high speed conditions
5. When the engine is at idling stage it mixes less amount of fuel to the air and sends to the engine
6. According to the load and speed of the engine the air-fuel mixture is sent to the engine.

9.5.2 Requirements of Carburettor

1. For the required engine and load the air-fuel mixture should be mixed correctly and sent it to the engine cylinder.
2. During cooling and heating conditions, carburettor should able to start the engine easily.

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